

Under the uniform class prior and the matched case $\beta = \kappa$, $q_\beta(y | x) = p(y | x)$, so

$$\begin{aligned}\mathbb{E}_x[q_\beta(0 | x) | y = 1] &= \int p(0 | x)p(x | y = 1) dx \\ &= 2 \int p(0 | x)p(1 | x)p(x) dx \\ &= 2\mathbb{E}_x[q_\beta(0 | x)q_\beta(1 | x)].\end{aligned}$$

Similarly,

$$\mathbb{E}_x[q_\beta(1 | x) | y = 0] = 2\mathbb{E}_x[q_\beta(0 | x)q_\beta(1 | x)].$$

Using these identities in (26), we obtain

$$\begin{aligned}\nabla_{\mathbf{v}}\mathcal{J}(\mathbf{v}) &= \beta \mathbb{E}_x[q_\beta(0 | x)q_\beta(1 | x)](\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top (\nabla_{\mathbf{v}}\boldsymbol{\mu}_1(\mathbf{v}) - \nabla_{\mathbf{v}}\boldsymbol{\mu}_0(\mathbf{v})) \\ &\quad + \frac{\beta}{2}(\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top (\Delta_1(\mathbf{v}) - \Delta_0(\mathbf{v})).\end{aligned}$$

By using

$$\nabla_{\mathbf{v}}\|\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v})\|^2 = 2(\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top (\nabla_{\mathbf{v}}\boldsymbol{\mu}_1(\mathbf{v}) - \nabla_{\mathbf{v}}\boldsymbol{\mu}_0(\mathbf{v})),$$

we obtain

$$\nabla_{\mathbf{v}}\mathcal{J}(\mathbf{v}) = \frac{\beta}{2} \mathbb{E}_x[q_\beta(0 | x)q_\beta(1 | x)] \nabla_{\mathbf{v}}\|\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v})\|^2 + \frac{\beta}{2}(\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top (\Delta_1(\mathbf{v}) - \Delta_0(\mathbf{v})).$$

Finally, using the logistic form of q_β ,

$$\begin{aligned}\frac{1}{2}\mathbb{E}_x[q_\beta(0 | x)q_\beta(1 | x)] &= \mathbb{E}_x\left[\frac{\exp\left\{\beta(\boldsymbol{\mu}_0(\mathbf{v}) + \boldsymbol{\mu}_1(\mathbf{v}))^\top f_{\theta, \mathbf{v}}(x)\right\}}{2\left(\sum_{c \in \{0,1\}} \exp\{\beta\boldsymbol{\mu}_c(\mathbf{v})^\top f_{\theta, \mathbf{v}}(x)\}\right)^2}\right] \\ &= \frac{1}{8}\left(1 - \mathbb{E}_x\left[\tanh^2\left(\frac{\beta}{2}(\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top f_{\theta, \mathbf{v}}(x)\right)\right]\right),\end{aligned}\tag{27}$$

which yields

$$\nabla_{\mathbf{v}}\mathcal{J}(\mathbf{v}) = \beta \bar{\omega}(\mathbf{v}) \nabla_{\mathbf{v}}\|\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v})\|^2 + \frac{\beta}{2}(\boldsymbol{\mu}_1(\mathbf{v}) - \boldsymbol{\mu}_0(\mathbf{v}))^\top (\Delta_1(\mathbf{v}) - \Delta_0(\mathbf{v})).$$

This completes the proof. $\square$

H.2 Distribution Shift on Two Types of Errors

In practice, the deployment distribution may differ from the source distribution used for training and calibration. We consider this mismatch as a distribution shift between the source and deployment distributions. In this section, we quantify its impact via a Wasserstein perturbation and study how it affects Type I error control.

Assumption 4. We denote by $\mathbb{P}_c$ the source distribution of class c used for training and calibration, and by $\tilde{\mathbb{P}}_c$ the corresponding class-conditional distribution under deployment. Assume there exists a constant $\mathcal{E} > 0$ such that, for a fixed representation map $f_{\theta, \mathbf{v}} : \mathcal{X} \rightarrow \mathbb{S}^{d-1}$, the following holds for each class $c \in \{0, 1\}$:

$$\mathcal{W}_1\left((f_{\theta, \mathbf{v}})_\# \mathbb{P}_c, (f_{\theta, \mathbf{v}})_\# \tilde{\mathbb{P}}_c\right) \leq \mathcal{E},$$

where $\mathcal{W}_1(\mu, \nu) := \inf_{\gamma \in \Pi(\mu, \nu)} \mathbb{E}_{(z, \tilde{z}) \sim \gamma} [\|z - \tilde{z}\|]$ denotes the 1-Wasserstein distance on $\mathbb{S}^{d-1}$, and $(f_{\theta, \mathbf{v}})_\# \mathbb{P}_c$ denotes the pushforward measure of $\mathbb{P}_c$ under $f_{\theta, \mathbf{v}}$.

Proposition H.2. Suppose Assumption 4 holds. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the randomness of training and calibration samples, the Type-I error under the shifted distribution $\tilde{\mathbb{P}}_0$ satisfies

$$\tilde{\mathbb{P}}_0(\hat{\mathcal{S}}_t(X) > \hat{\tau}_{\alpha, t}) - \alpha \leq \inf_{\varepsilon > 0} \left(\sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2} + \mathbb{P}_0(\hat{\tau}_{\alpha, t} - \varepsilon \leq \hat{\mathcal{S}}_t(X) \leq \hat{\tau}_{\alpha, t}) + \frac{2\kappa\mathcal{E}}{\varepsilon} \right) \wedge 1.$$

The Type-II error under the shifted distribution $\tilde{\mathbb{P}}_1$ satisfies

$$\tilde{\mathbb{P}}_1(\hat{\mathcal{S}}_t(X) \leq \hat{\tau}_{\alpha, t}) \leq \inf_{\varepsilon > 0} \left(\mathbb{P}_1(\hat{\mathcal{S}}_t(X) \leq \hat{\tau}_{\alpha, t} + \varepsilon) + \frac{2\kappa\mathcal{E}}{\varepsilon} \right) \wedge 1.$$

Proposition H.2 characterizes how the Type-I and Type-II errors depend on the magnitude of the distribution shift $\mathcal{E}$. When $\mathcal{E} = 0$, the bounds recover the in-distribution guarantees in Theorem 3.1. When $\mathcal{E} > 0$, both errors incur an additional penalty that scales with $\mathcal{E}$, reflecting performance degradation under shift. The parameter ε captures a trade-off between the local probability mass near the threshold and the shift-induced error. Overall, the result indicates that the proposed method is robust to moderate shifts, while its performance degrades gracefully as the distribution shift increases.

Proof of Proposition H.2. For simplicity of notation, we drop the time index t in the proof. Let $f_{\theta, \mathbf{v}}(x) \in \mathbb{S}^{d-1}$ denote the representation for $x \in \mathcal{X}$. The deployed scoring function is defined as $\hat{\mathcal{S}}(x) = \kappa(\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^\top f_{\theta, \mathbf{v}}(x)$. Since the estimated mean directions $\hat{\boldsymbol{\mu}}_1, \hat{\boldsymbol{\mu}}_0$ lie on the unit hypersphere $\mathbb{S}^{d-1}$, by the triangle inequality, $\|\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0\| \leq \|\hat{\boldsymbol{\mu}}_1\| + \|\hat{\boldsymbol{\mu}}_0\| \leq 2$. The gradient of $\hat{\mathcal{S}}$ with respect to the representation f is given by

$$\|\nabla_f \hat{\mathcal{S}}(x)\| = \kappa \|\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0\| \leq 2\kappa. \quad (28)$$

Let $\pi \in \Pi(\mathbb{P}_0, \tilde{\mathbb{P}}_0)$ be the optimal coupling such that

$$\mathbb{E}_{(X, \tilde{X}) \sim \pi} [\|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\|] = \mathcal{W}_1((f_{\theta, \mathbf{v}})_\# \mathbb{P}_0, (f_{\theta, \mathbf{v}})_\# \tilde{\mathbb{P}}_0) \leq \mathcal{E}.$$

For any $\varepsilon > 0$, we decompose the Type-I error probability under the shifted distribution as

$$\begin{aligned} \tilde{\mathbb{P}}_0(\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha) &\leq \mathbb{P}_\pi(\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha, \|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| \leq \frac{\varepsilon}{2\kappa}) \\ &\quad + \mathbb{P}_\pi(\|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| > \frac{\varepsilon}{2\kappa}). \end{aligned} \quad (29)$$

For the first term on the RHS of (29), by using (28), we obtain

$$|\hat{\mathcal{S}}(X) - \hat{\mathcal{S}}(\tilde{X})| \leq 2\kappa \|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| \leq \varepsilon.$$

Thus, $\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha$ implies $\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha - \varepsilon$. By Theorem 3.1, with probability at least $1 - \delta$ over the calibration sample $\mathcal{S}_{\text{cal}}$,

$$\mathbb{P}_0(\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha) \leq \alpha + \sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2}.$$

On this high-probability event, we therefore have

$$\mathbb{P}_\pi(\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha, \|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| \leq \frac{\varepsilon}{2\kappa}) \leq \mathbb{P}_0(\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha) + \mathbb{P}_0(\hat{\tau}_\alpha - \varepsilon \leq \hat{\mathcal{S}}(X) \leq \hat{\tau}_\alpha). \quad (30)$$

For the second term on the RHS of (29), applying Markov's inequality yields

$$\mathbb{P}_\pi(\|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| > \frac{\varepsilon}{2\kappa}) \leq \frac{2\kappa}{\varepsilon} \mathbb{E}_\pi[\|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\|] \leq \frac{2\kappa\mathcal{E}}{\varepsilon}.$$

Substituting (30) into (29) and applying Markov's inequality, we obtain, for any $\varepsilon > 0$,

$$\begin{aligned} \tilde{\mathbb{P}}_0(\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha) &\leq \mathbb{P}_0(\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha) + \mathbb{P}_0(\hat{\tau}_\alpha - \varepsilon \leq \hat{\mathcal{S}}(X) \leq \hat{\tau}_\alpha) + \frac{2\kappa\mathcal{E}}{\varepsilon} \\ &\leq \alpha + \sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2} + \mathbb{P}_0(\hat{\tau}_\alpha - \varepsilon \leq \hat{\mathcal{S}}(X) \leq \hat{\tau}_\alpha) + \frac{2\kappa\mathcal{E}}{\varepsilon}. \end{aligned}$$

Taking the infimum over $\varepsilon > 0$ gives the stated Type-I bound.

Now we turn to the Type-II error. Let $\pi' \in \Pi(\mathbb{P}_1, \tilde{\mathbb{P}}_1)$ be an optimal coupling satisfying

$$\mathbb{E}_{(X, \tilde{X}) \sim \pi'} [\|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\|] \leq \mathcal{E}.$$

For any $\varepsilon > 0$,

$$\begin{aligned} \tilde{\mathbb{P}}_1(\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha) &\geq \mathbb{P}_{\pi'}(\hat{\mathcal{S}}(\tilde{X}) > \hat{\tau}_\alpha, \|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| \leq \frac{\varepsilon}{2\kappa}) \\ &\geq \mathbb{P}_{\pi'}(\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha + \varepsilon, \|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| \leq \frac{\varepsilon}{2\kappa}) \\ &\geq \mathbb{P}_1(\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha + \varepsilon) - \mathbb{P}_{\pi'}(\|f_{\theta, \mathbf{v}}(X) - f_{\theta, \mathbf{v}}(\tilde{X})\| > \frac{\varepsilon}{2\kappa}) \\ &\geq \mathbb{P}_1(\hat{\mathcal{S}}(X) > \hat{\tau}_\alpha + \varepsilon) - \frac{2\kappa\mathcal{E}}{\varepsilon}. \end{aligned}$$

Equivalently,

$$\tilde{\mathbb{P}}_1(\hat{\mathcal{S}}(\tilde{X}) \leq \hat{\tau}_\alpha) \leq \mathbb{P}_1(\hat{\mathcal{S}}(X) \leq \hat{\tau}_\alpha + \varepsilon) + \frac{2\kappa\mathcal{E}}{\varepsilon}.$$

Taking the infimum over $\varepsilon > 0$ gives the stated Type-II bound. $\square$

I Broader Impacts

The proposed S2D framework contributes to improving the reliability of distinguishing LLM-generated text from human-written text, which has implications for mitigating risks such as misinformation, academic misconduct, and erosion of trust in digital content. By leveraging hidden representations and providing a statistically grounded detection procedure with controlled Type-I error, our method is particularly relevant in high-stakes scenarios where false accusations (e.g., misclassifying human-written text as machine-generated) carry ethical consequences. In addition, the representation-based nature of S2D allows it to operate without requiring access to the generation process, making it broadly applicable in real-world settings where watermarking is unavailable.

More broadly, such capabilities may influence how content is created and consumed, potentially shaping emerging norms around disclosure and authenticity. They may also inform policy discussions on transparency and responsible AI use, including whether and how generated content should be labeled. Overall, advances in this area contribute to a broader effort to maintain trust, accountability, and informed decision-making in an increasingly AI-mediated information landscape.